

UNIT III

Fluid Kinematics

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FLUID

A matter has three states – Solid and fluid (Liquid and gas).

A substance which can flow freely when not in equilibrium condition is called fluid.

Fluid has the following characteristics;

1. It has no specific shape of its own, it confirms the shape of containing vessel.
2. A small amount of shear force exerted on fluid will cause it to deform and continue to deform as long as the force exist.

PROPERTIES OF LIQUID³

The properties of water are of much importance because the subject of hydraulics is mainly concerned with it.

Some important properties of liquid which will be considered are:

- (i) **Density – Mass or weight per unit volume (kg/m^3)**
- (ii) **Specific gravity – Ratio of specific weight of liquid to Sp. Wt. of Std. Liquid**
- (iii) **Viscosity – Resistance to shear stress between consecutive layers (N-s/m^2)**
- (iv) **Cohesion – Tendency of liquid to remain one assemblage of particles**

- (v) Adhesion – Attraction between molecules of liquids and solid boundary and liquid molecules**
- (vi) Surface tension – Caused by force of cohesion at the free surface**
- (vii) Capillarity – Rise of liquid into a glass tube due combined effect of cohesion and adhesion**
- (viii) Compressibility – Change in volume due to the external force**

VISCOSITY (NEWTONS LAW OF VISCOSITY) ⁵

When two layers of fluid, at a distance ' dy ' apart, move one over the other at different velocities, say u and $u + du$, the viscosity together with relative velocity causes a shear stress acting between the fluid layers. The top layer causes a shear stress on the adjacent lower layer while the lower layer causes a shear stress on the adjacent top layer.

Mathematically

$$\tau \propto \frac{du}{dy}$$

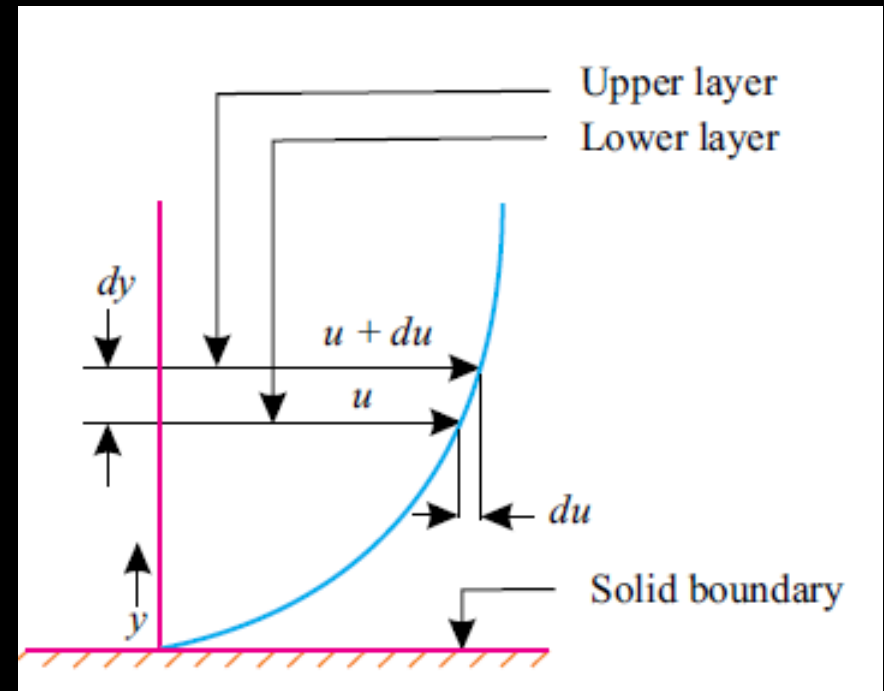
$$\tau = \mu \cdot \frac{du}{dy}$$

$$\frac{du}{dy}$$

= Velocity Gradient or Rate of shear deformation

$$\mu$$

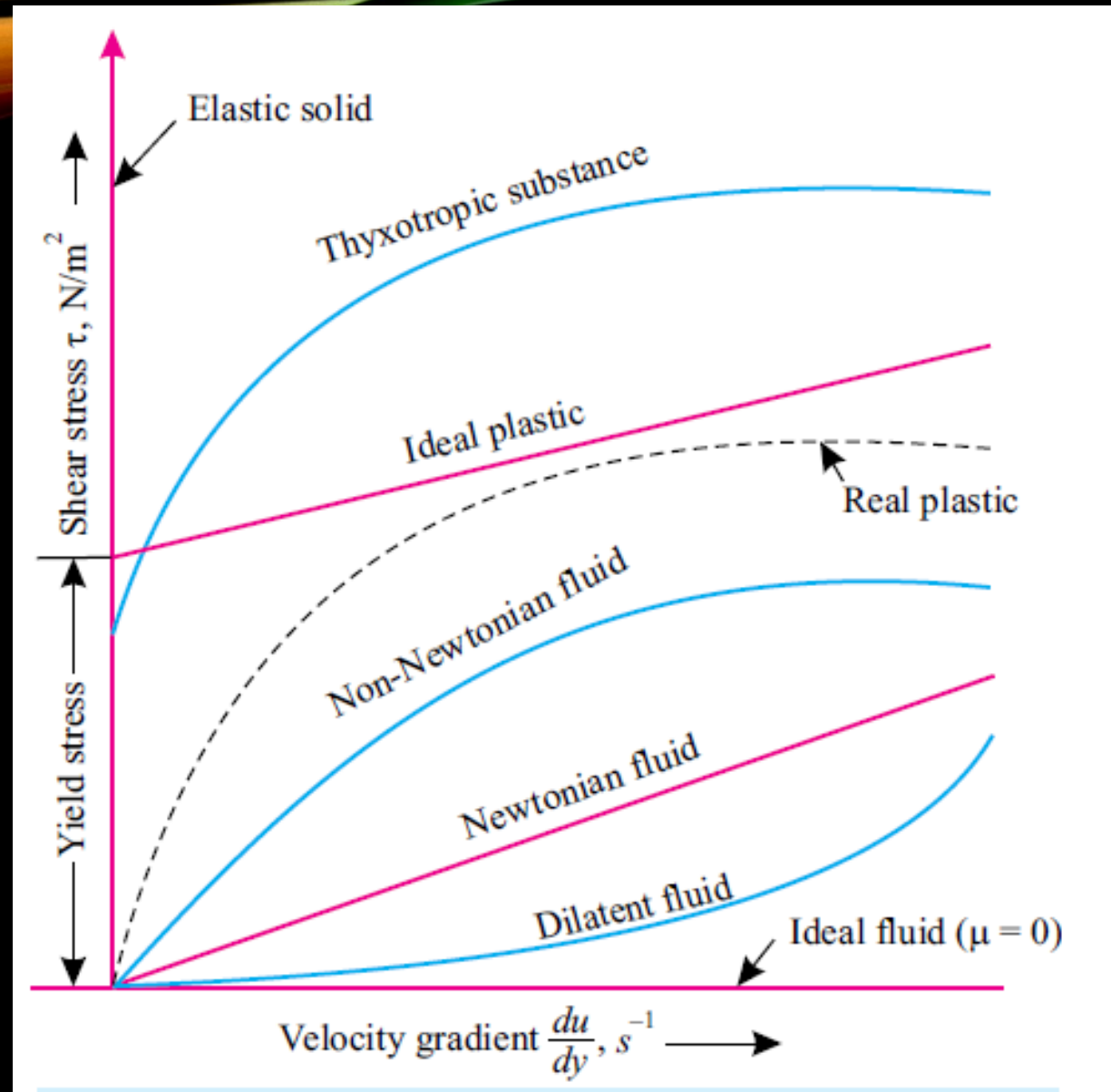
= Viscosity



TYPES OF FLUIDS⁶

According to **Newton's Law of viscosity**, the shear stress is proportional to the rate of change of velocity with respect to y . It is denoted by τ .

1. **Newtonian fluids** These fluids follow Newton's viscosity equation. For such fluids μ does not change with rate of deformation. **Examples** - Water, kerosene, air etc.
2. **Non-Newtonian** Fluids which do not follow the linear relationship between shear stress and rate of deformation are termed as Non-Newtonian fluids. Such fluids are relatively uncommon. **Examples**- Solutions or suspensions (slurries), mud flows, polymer solutions, blood etc. These fluids are generally complex mixtures and are studied under rheology, a science of deformation and flow.



KINEMATICS OF FLUID FLOWS

Kinematics means study of motion therefore fluid kinematics is the study of fluid motion or describing the motion of fluid flow without considering the forces those are responsible for flow.

The fluid flow always caused by some external force, gravity etc. Displacement, velocity and acceleration is the focus of study.

Study of fluid motion is conducted in two different ways.

Motion of each individual particle is tracked to study its displacement, velocity and acceleration. This method is Lagrangian method.

In the other method a field variable at selected location is studied over a period of time. This method is Eulerian method

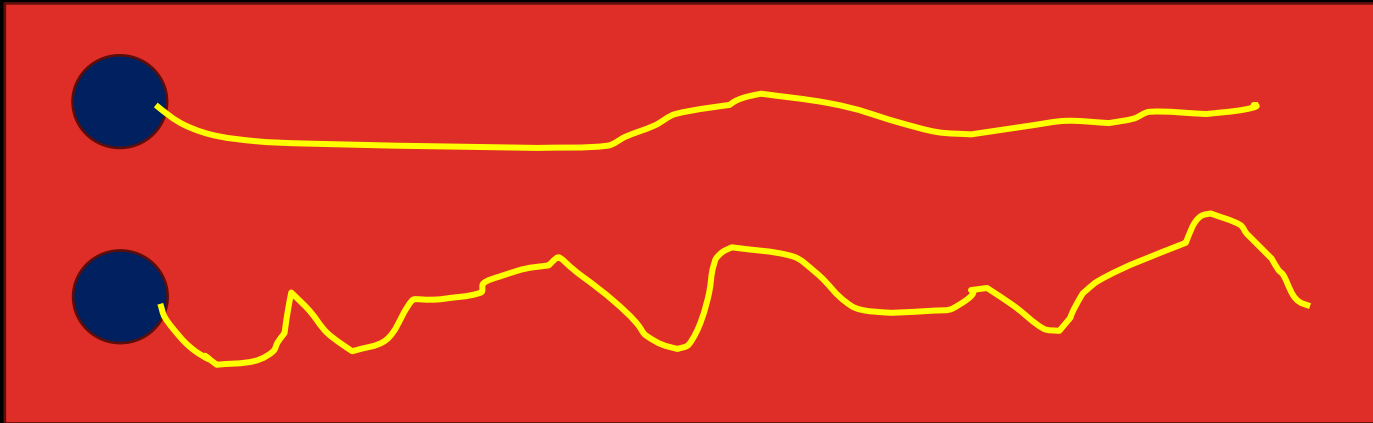
LAGRANGIAN METHOD

Lagrangian Method (Fluid Coordinates)

The motion of the fluid particles is described by specifying their coordinates as a function of time. The line connecting all points a particle is passing through in the course of time is called path line or Lagrangian particle path.

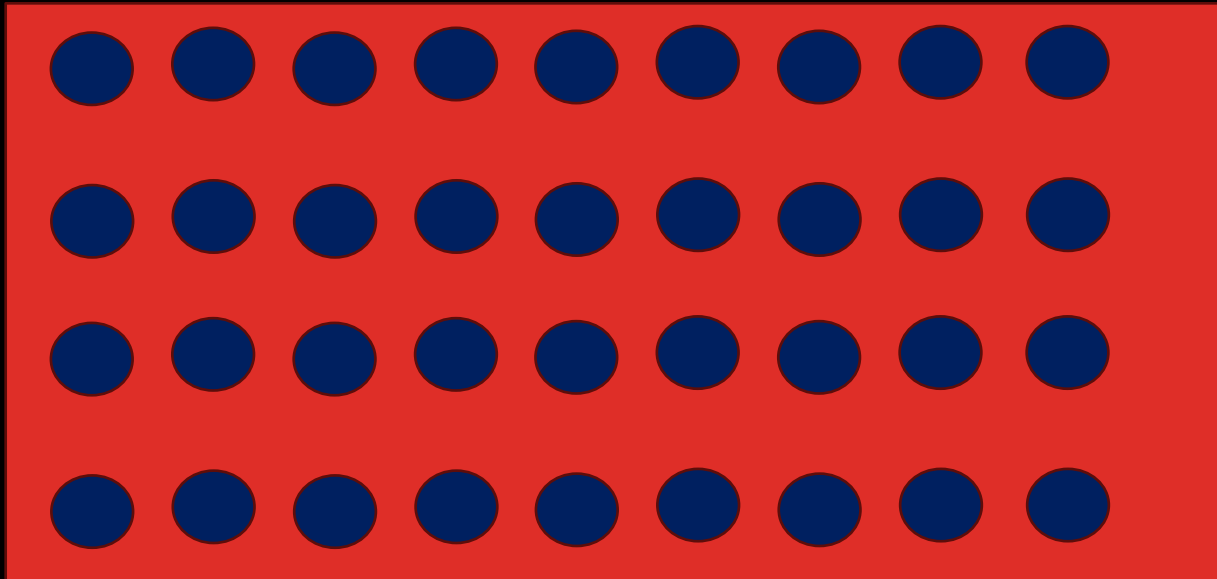
The path line begins at time $t = t_0$ at a point defined by the position vector

$$\mathbf{r}_0 = i x_0 + j y_0 + k z_0$$



LAGRANGIAN METHOD

- In actual systems there are a large number of fluid particles
- Tracking each and every particle becomes time consuming job
- Days or even months are required to for small quantity of fluid to conduct study
- This method becomes a laborious and time consuming for kinematic study of fluid



The motion of the fluid is completely described, if the position vector $\mathbf{r}$ is known as a function of time:

$$\mathbf{r} = \mathbf{F}(\mathbf{r}_0, t)$$

or in components

$$x = f_1(x_0, y_0, z_0, t) ;$$

$$y = f_2(x_0, y_0, z_0, t) ;$$

$$z = f_3(x_0, y_0, z_0, t) .$$

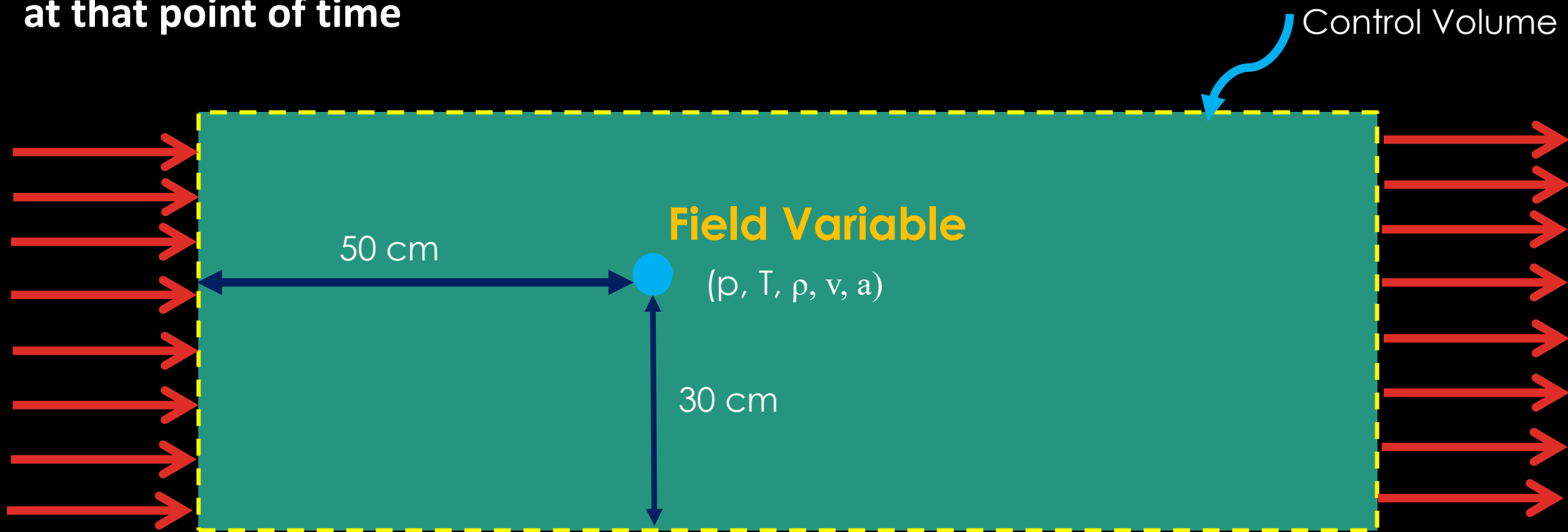
The velocity is obtained by differentiating the position vector with respect to time

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \left(\frac{\mathbf{r}_2 - \mathbf{r}_1}{\Delta t} \right) = \frac{d\mathbf{r}}{dt}$$

EULARIAN METHOD

Eulerian Method (Space Coordinates)

- In this method a point of interest is selected at which a field variable is defined
- The field variable may be pressure, temperature, velocity, acceleration or any property that defines the fluid.
- Field variables are referred to any fluid particle occupies the selected location at that point of time



EULARIAN METHOD

The field variable is function of space and time

Field variable = function(space, time)

Let take velocity a field variable

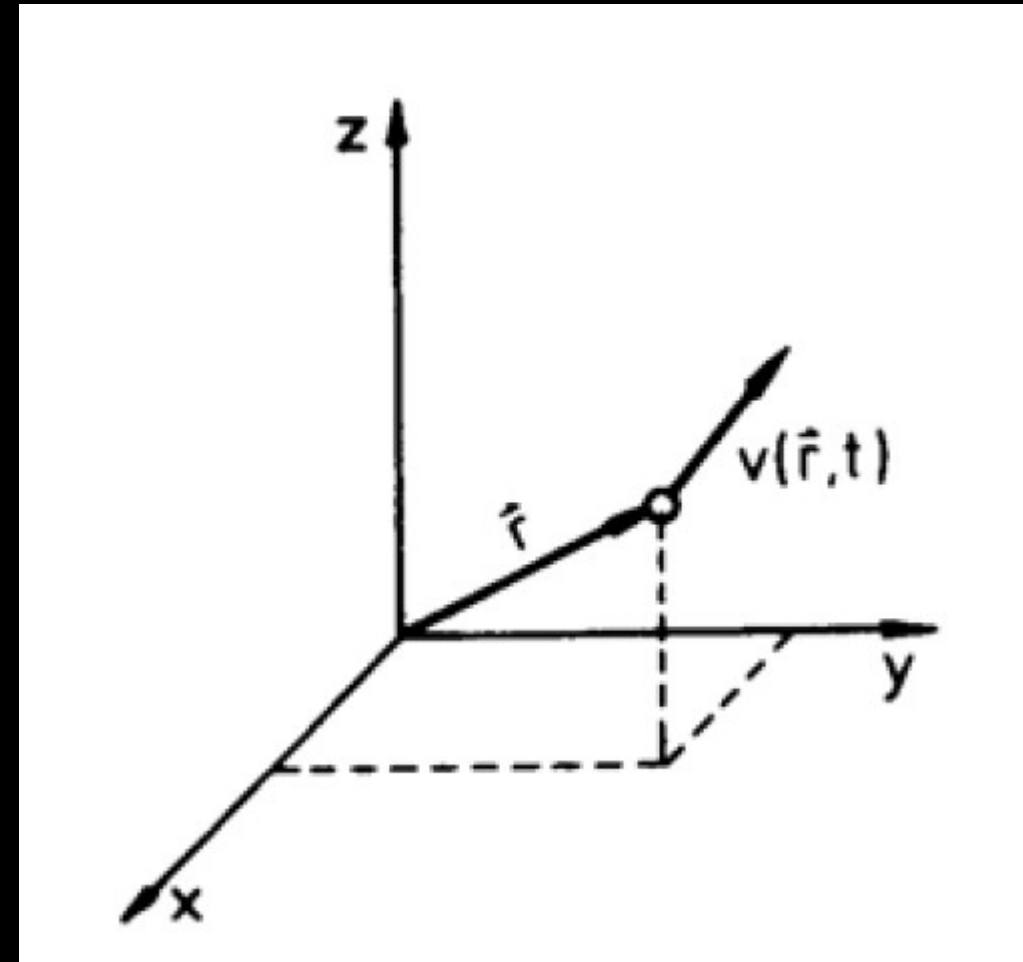
Velocity = function(x, y, z, t)

or written in components

$u = g1(x, y, z, t) ;$

$v = g2(x, y, z, t) ;$

$w = g3(x, y, z, t)$



Example: In a fluid, the velocity field is given by

$$\mathbf{V} = (3x + 2y) \mathbf{i} + (2z + 3x^2) \mathbf{j} + (2t - 3z) \mathbf{k}$$

Determine:

- (i) The velocity components u , v , w at any point in the flow field;
- (ii) The speed at point $(1, 1, 1)$;
- (iii) The speed at time $t = 2\text{s}$ at point $(0, 0, 2)$. Also classify the velocity field as steady, or unsteady, uniform or non-uniform and one, two or three dimensional.

Solution.

Given: Velocity field, $\mathbf{V} = (3x + 2y) \mathbf{i} + (2z + 3x^2) \mathbf{j} + (2t - 3z) \mathbf{k}$

- (i) Velocity components: The velocity components are:

$$u = 3x + 2y, \quad v = (2z + 3x^2), \quad w = (2t - 3z)$$

(ii) Speed at point $(1, 1, 1)$, $V_{(1,1,1)}$;

Substituting $x = 1, y = 1, z = 1$ in the expressions for u, v and w ,

we have:

$$u = (3+2) = 5,$$

$$v = (2+3) = 5,$$

$$w = (2t-3)$$

$$\begin{aligned} \text{Velocity field, } V^2 &= u^2 + v^2 + w^2 \\ &= 5^2 + 5^2 + (2t-3)^2 \\ &= 25 + 25 + 4t^2 - 6t + 9 \\ V^2 &= 4t^2 - 6t + 49 \\ V &= \sqrt{4t^2 - 6t + 49} \end{aligned}$$

(iii) Speed at $t = 2s$ at point $(0, 0, 2)$:

Substituting $t = 2, x=0, y=0, z=2$ in the expressions for u, v and w , we get:

$$u = 0,$$

$$v = (2 \times 2 + 0) = 4,$$

$$w = (2 \times 2 - 3 \times 2) = -2$$

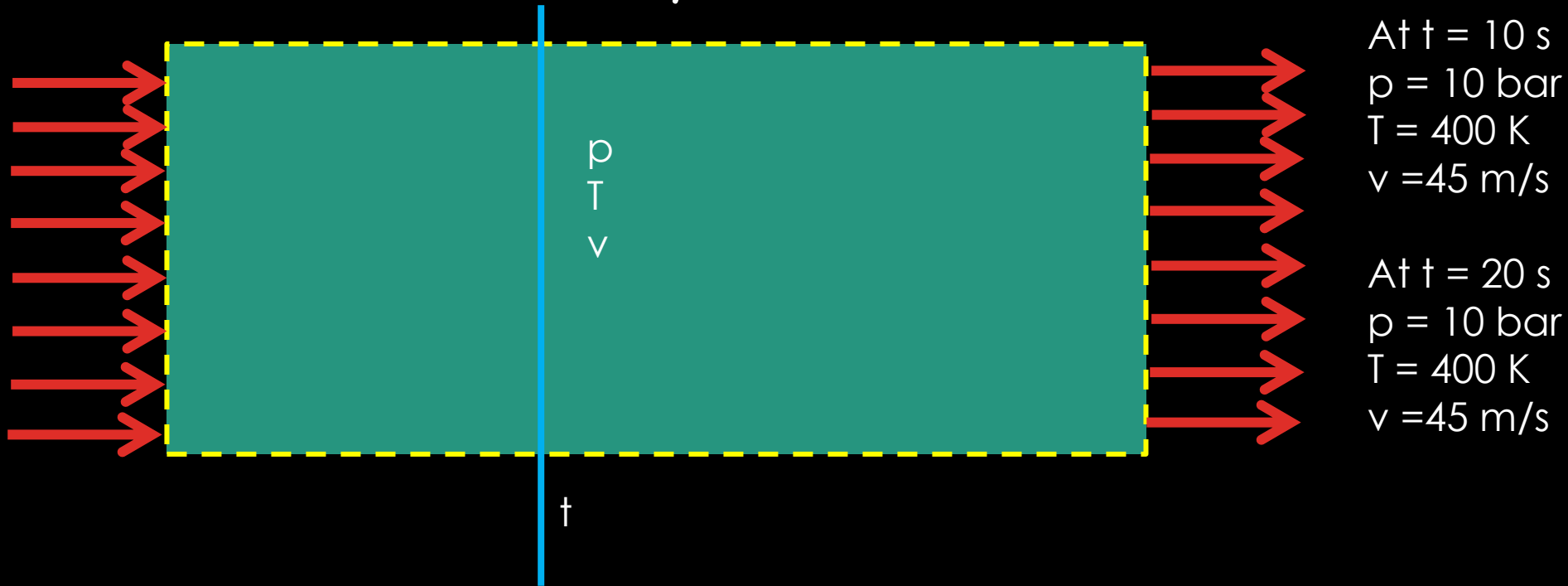
$$V^2 = u^2 + v^2 + w^2 = 0^2 + 4^2 + (-2)^2 = 20$$

$$V_{(2,0,2)} = \sqrt{20} = 4.472$$

1. Steady and Unsteady Flow

Steady Flow

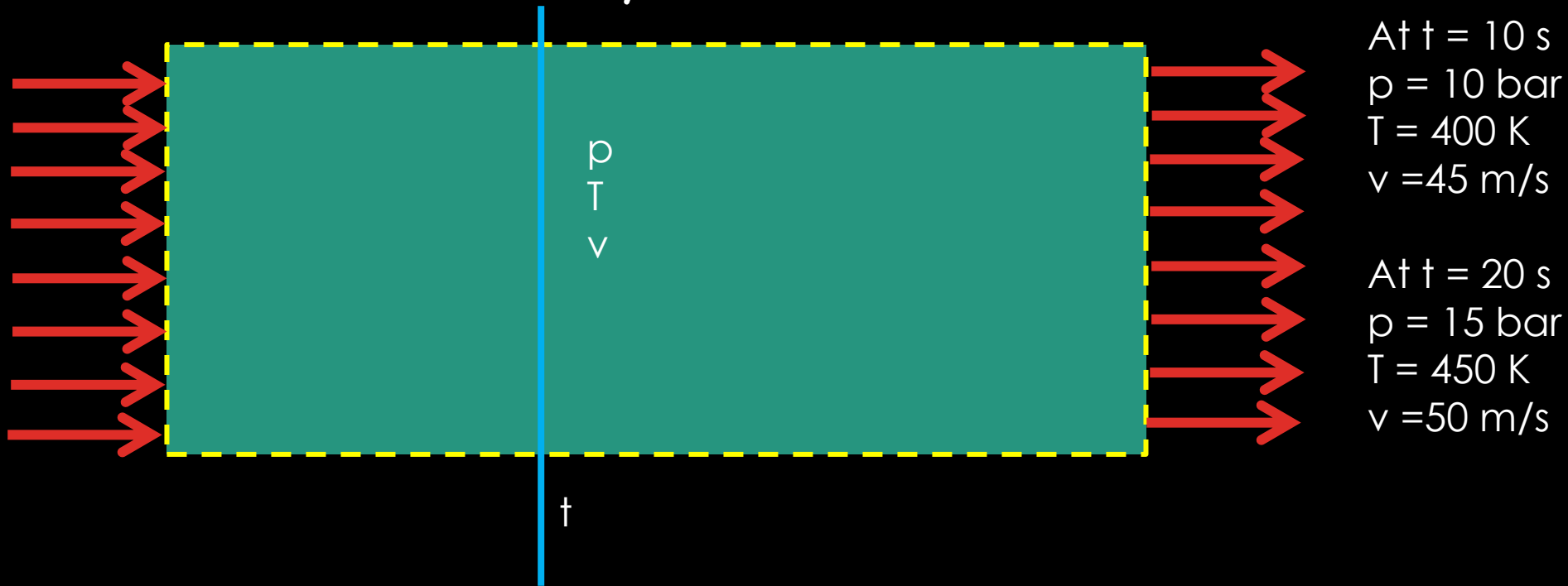
When the pressure, temperature, velocity etc. do not change with time the flow is known as steady flow



1. Steady and Unsteady Flow

Unsteady Flow

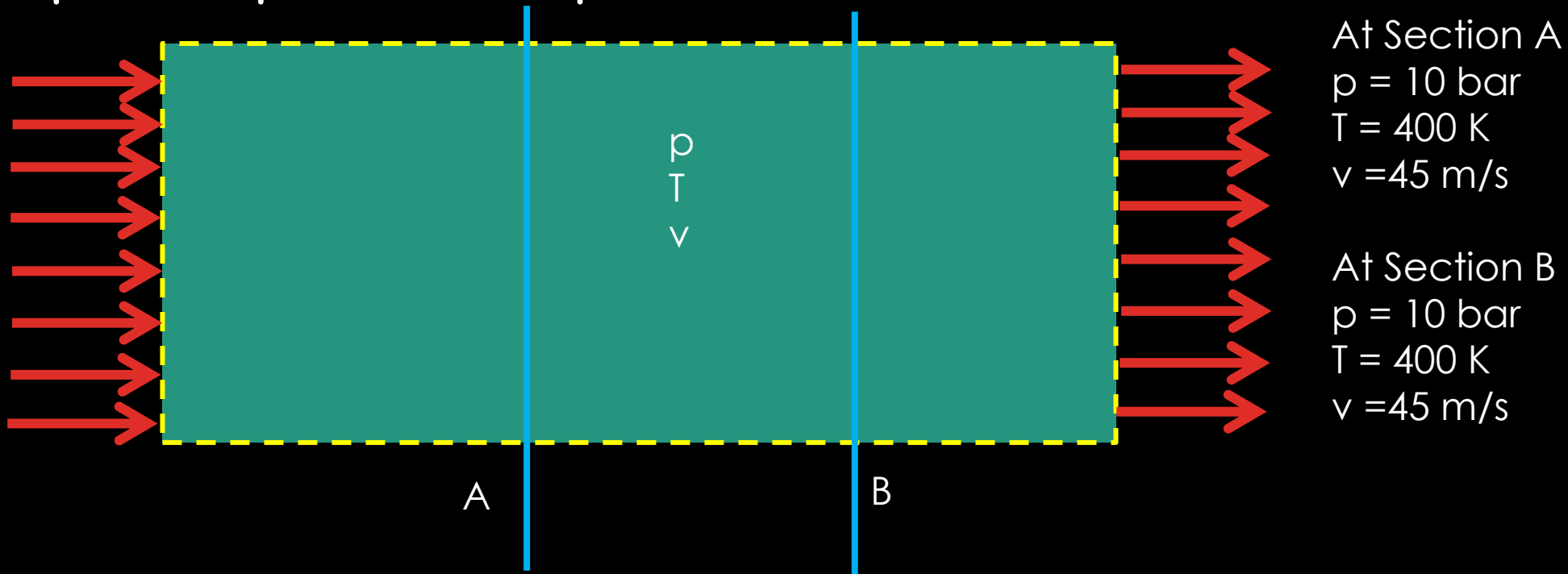
When the pressure, temperature, velocity etc. changes with time the flow is known as unsteady flow



2. Uniform and Non-Uniform Flow

Uniform Flow

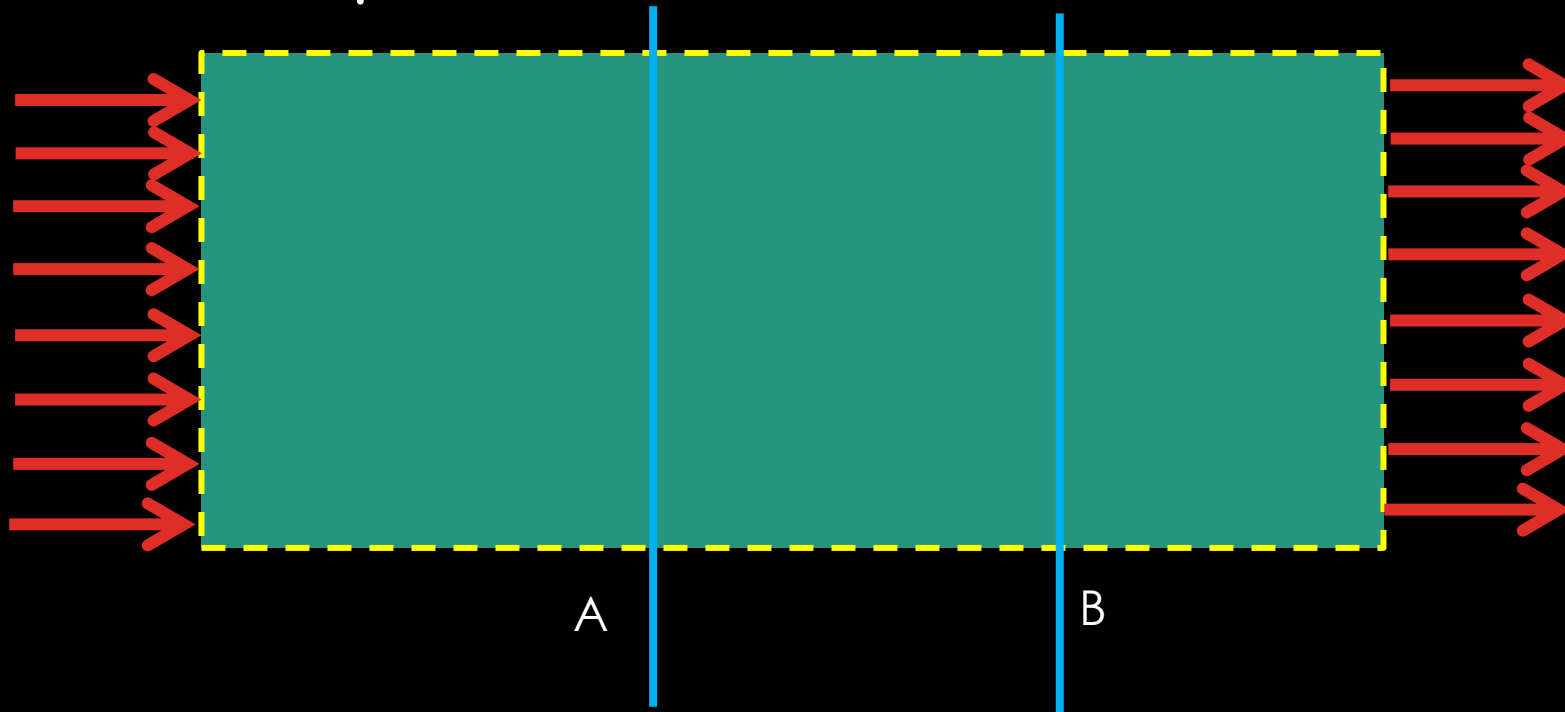
When the pressure, temperature, velocity etc. do not change with respect to position or space the is known as uniform flow



2. Uniform and Non-Uniform Flow

Non-Uniform Flow

When the pressure, temperature, velocity etc. varying with respect to position or space the is known as non-uniform flow



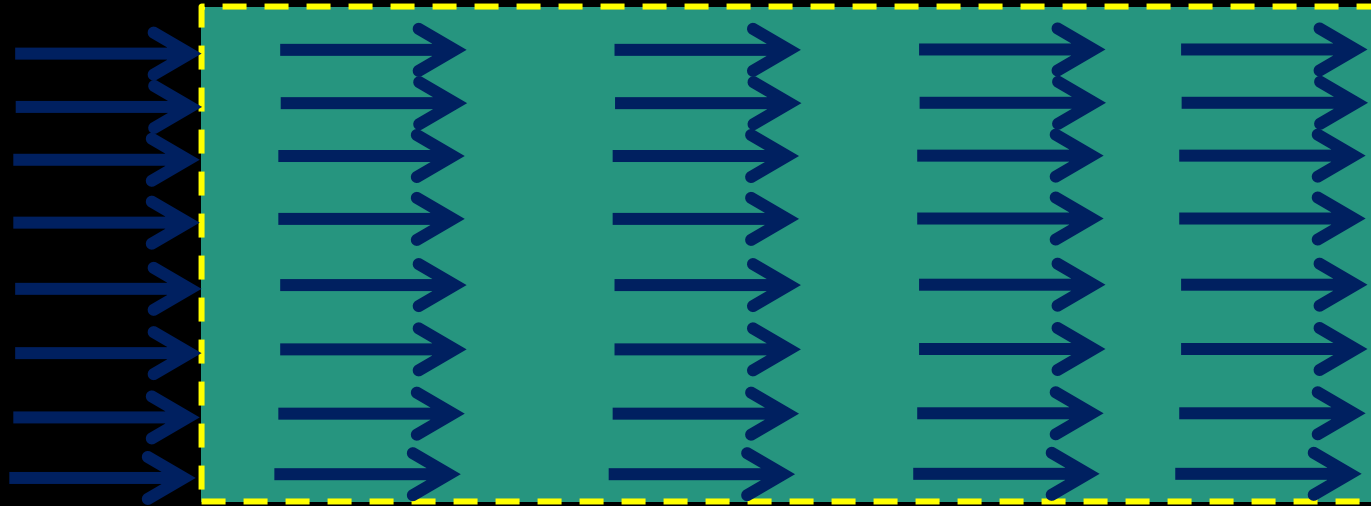
At Section A
 $p = 10 \text{ bar}$
 $T = 400 \text{ K}$
 $v = 45 \text{ m/s}$

At Section B
 $p = 15 \text{ bar}$
 $T = 500 \text{ K}$
 $v = 70 \text{ m/s}$

3. Laminar and Turbulent Flow

Laminar Flow

When the fluid particle flows in a defined straight and parallel path the flow is known as laminar flow

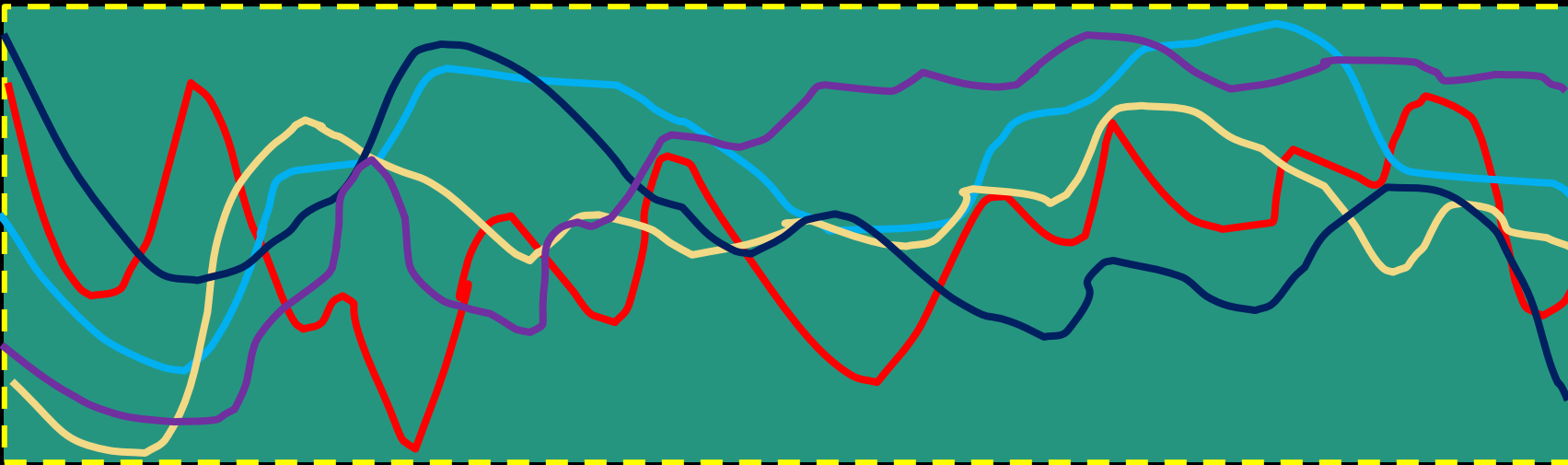


Example:
March past of Army

3. Laminar and Turbulent Flow

Turbulent Flow

When the fluid particle flows in a random fashion and no specific path the flow is known as turbulent flow



Example:

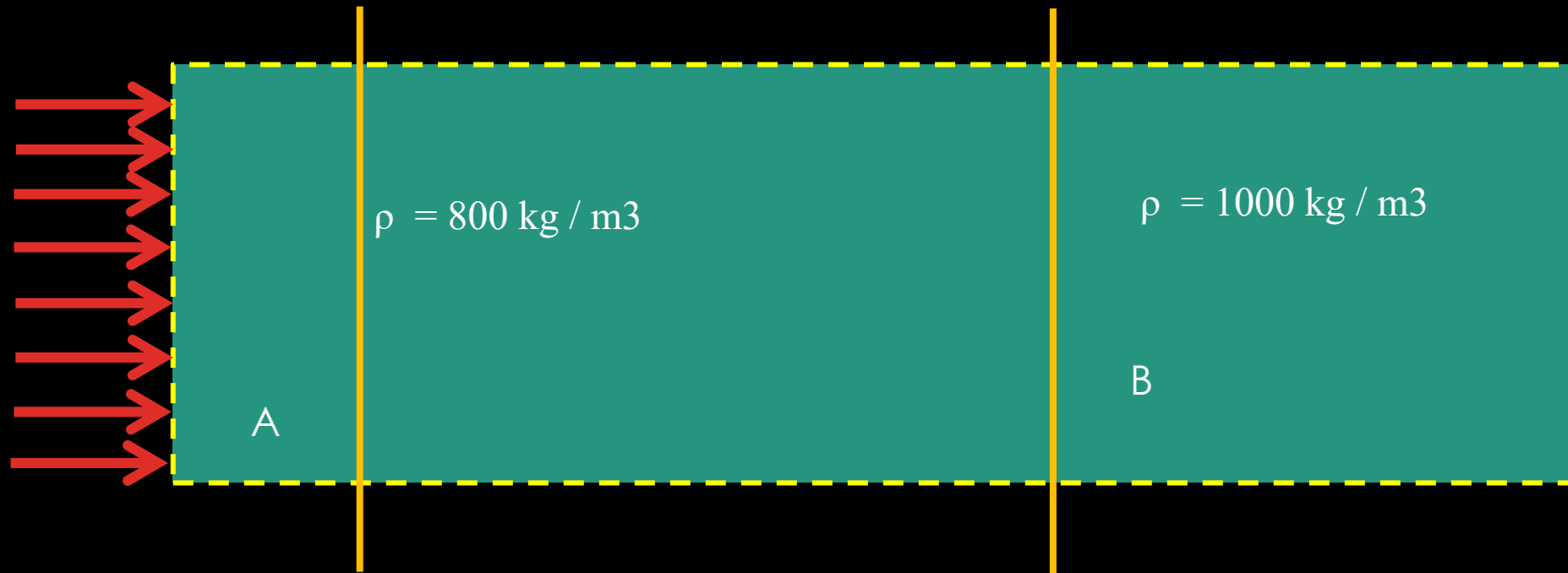
Passengers at railway platform

People moving in shopping mall

4. Compressible and Incompressible Flow

Compressible Flow

When the fluid density is varying from point to point the flow is known as compressible flow

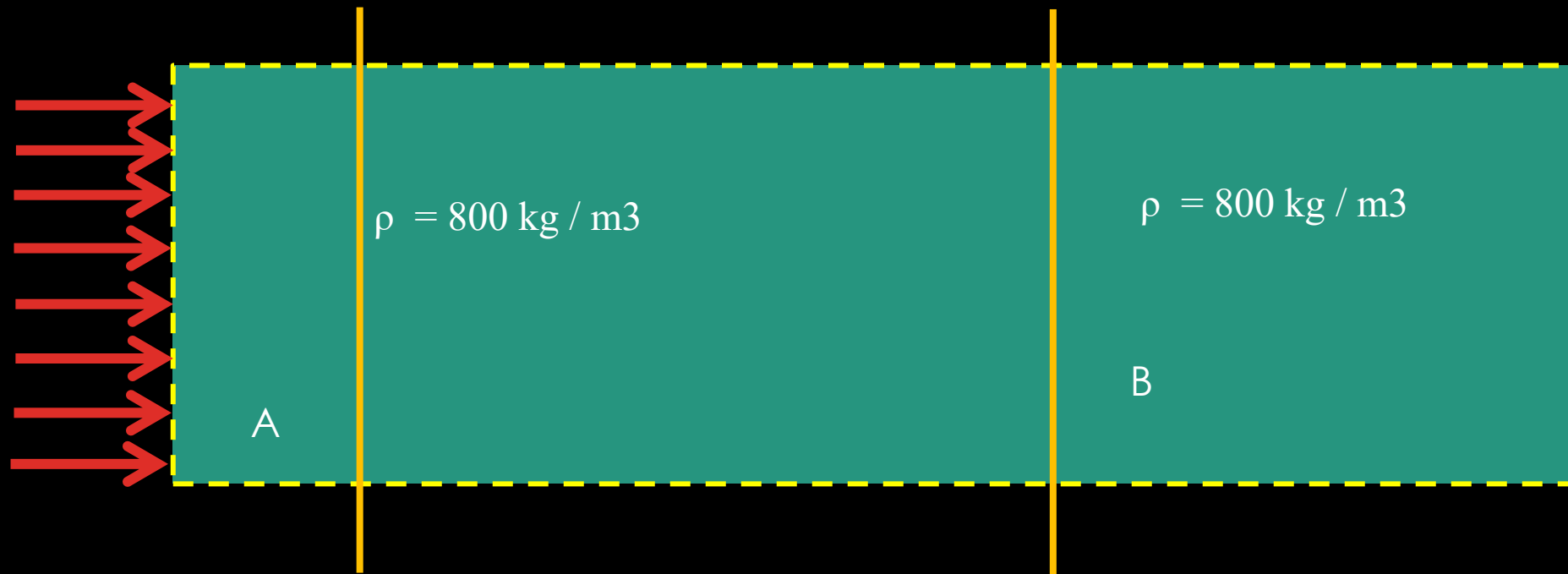


e.g. Flow of air through compressor

4. Compressible and Incompressible Flow

Incompressible Flow

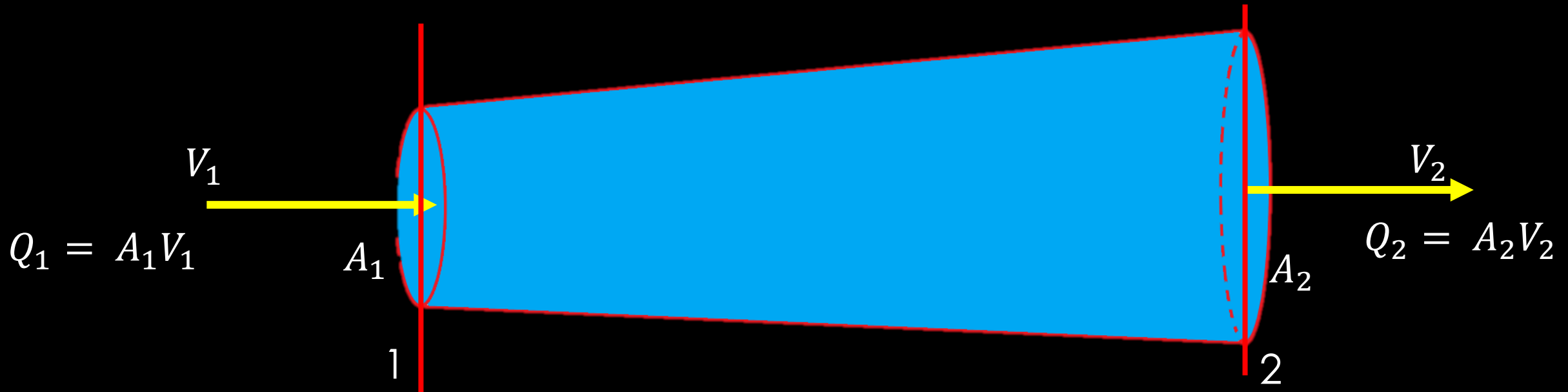
When the fluid density is varying from point to point the flow is known as incompressible flow



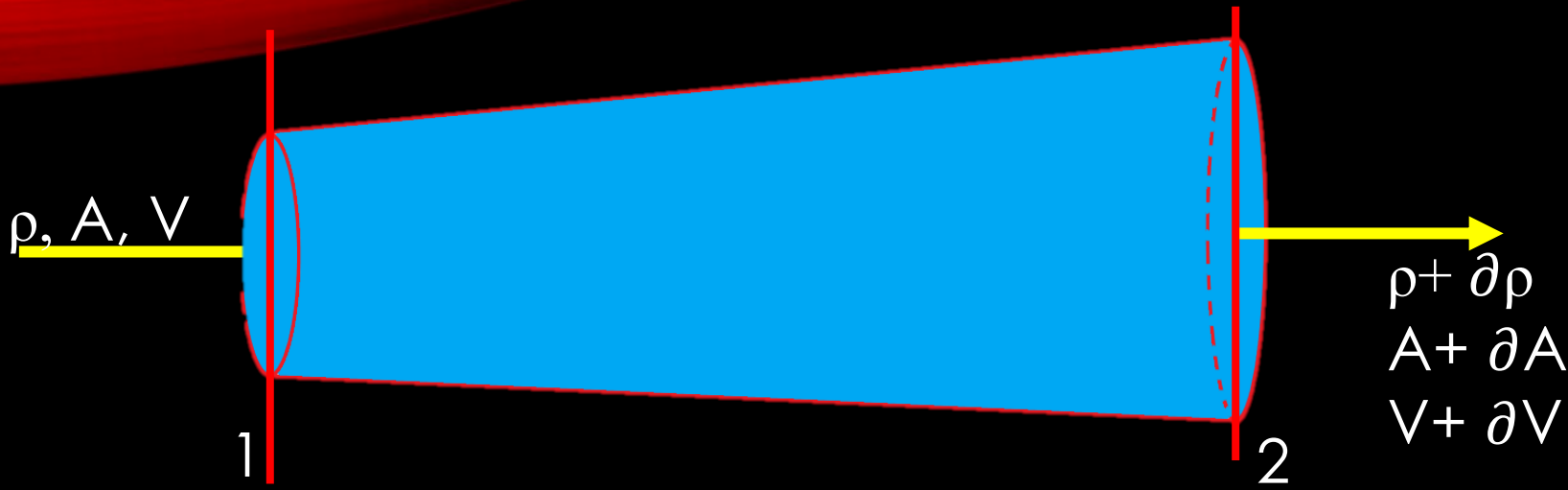
CONTINUITY EQUATION²⁵

Discharge

- Discharge is defined as the volume of fluid flowing through a section of pipe or a channel per unit time.
- It is represented by letter 'Q' or ' $\dot{V}$ ' it is representing flow rate
- Unit of discharge is cubic meter per m^3/s second or liter per second *litert/s*



CONTINUITY EQUATION ²⁶



ρ = Density

A = Cross section Area

V = Velocity

Mass flow rate of fluid entering the section 1 = ρAV

Mass flow rate of fluid leaving the section 2 = $(\rho + \partial\rho)(A + \partial A)(V + \partial V)$

$[\rho = \text{mass/volume} = \text{mass flow rate} / \text{vol flow rate}, \text{Mass flow rate} = \text{vol flow rate} \times \rho]$

CONTINUITY EQUATION ²⁷

Quantity of Mass accumulated between section 1 and section 2 per unit time = $\frac{dm}{dt}$

$\frac{dm}{dt}$ = mass of fluid entering the section 1 – mass of fluid of leaving the section 2

$$\frac{dm}{dt} = \rho AV - (\rho + \partial\rho)(A + \partial A)(V + \partial V)$$

$$\frac{dm}{dt} = \rho AV - [\rho AV + \rho A \partial V + AV \partial \rho + \rho V \partial A + \rho \partial A \partial V + A \partial V \partial \rho + V \partial A \partial \rho + \partial A \partial V \partial \rho]$$

$$\frac{dm}{dt} = -[\rho A \partial V + AV \partial \rho + \rho V \partial A]$$

CONTINUITY EQUATION ²⁸

For steady flow $\frac{dm}{dt} = 0$

$$\frac{dm}{dt} = 0,$$

$$0 = -[\rho A \partial V + AV \partial \rho + \rho V \partial A]$$

$$\rho A \partial V + AV \partial \rho + \rho V \partial A = 0$$

In partial differentiation form

$$\partial(\rho AV) = 0$$

Integrating the above equation, we get

$$\rho AV = \text{Constant}$$

Mass flow rate for steady flow is constant for any section

CONTINUITY EQUATION²⁹

Mass flow rate at section 1 = mass flow rate for section 2

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 = \text{Constant}$$

For a incompressible flow $\rho_1 = \rho_2$

$$A_1 V_1 = A_2 V_2$$

The above equation is continuity equation.

Discharge of flow through a pipe section remains constant for **steady** and **incompressible** flow

CONTINUITY EQUATION³⁰

The diameters of a pipe at the sections 1-1 and 2-2 are 200 mm and 300 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 4 m/s, find:

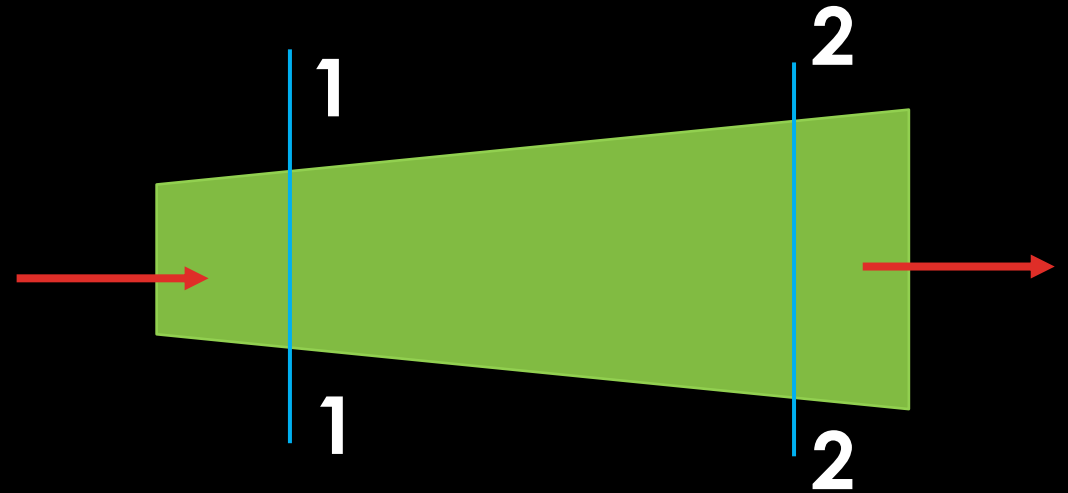
- (i) Discharge through the pipe, and
- (ii) Velocity of water at section 2-2

Solution:

Diameter of the pipe at section 1-1,
 $D_1 = 200 \text{ mm} = 0.2 \text{ m}$

$$\text{Area, } A_1 = \frac{\pi}{4} D_1^2 = \frac{\pi}{4} 0.2^2 = 0.0314 \text{ m}^2$$

$$\text{Velocity } V_1 = 4 \text{ m/s}$$



CONTINUITY EQUATION ³¹

Diameter of the pipe at section 2-2,

$$D_2 = 300 \text{ mm} = 0.3 \text{ m}$$

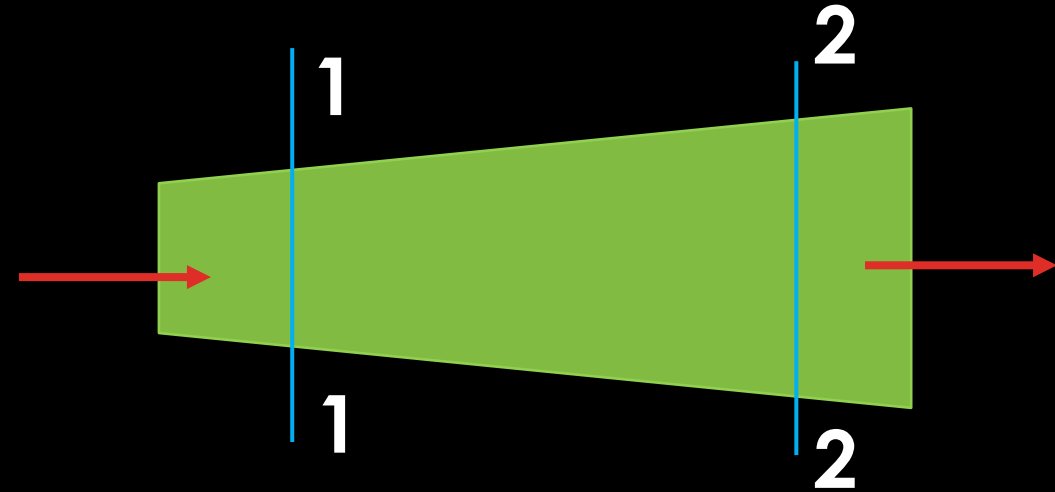
$$\text{Area, } A_2 = \frac{\pi}{4} D_2^2 = \frac{\pi}{4} 0.3^2 = 0.00707 \text{ m}^2$$

(i) Discharge through the pipe

$$Q = A_1 V_1 = 0.0314 \times 4 = 0.1256 \text{ m}^3 / \text{s}$$

(ii) Velocity of water at section 2-2

$$\text{Velocity } V_2 = \frac{A_1 V_1}{A_2} = \frac{0.0314 \times 4}{0.00707} = 1.77 \text{ m/s}$$





THE END

